

ST MICHAEL, AK, USA

WMO: 720382

Lat: 63.490N

Lon: 162.110W

Elev: 28

StdP: 100.99

Time Zone: -9.00 (NAK)

Period: 06-19

WBAN: 00123

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -30.9	(c) -27.8	(d) -33.0	(e) 0.2	(f) -30.3	(g) -31.2	(h) 0.2	(i) -27.8	(j) 17.2	(k) -9.7	(l) 15.5	(m) -8.5	(n) 3.0	(o) 230	(p) 0.857

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 5.7	(c) 19.5	(d) 15.0	(e) 18.0	(f) 14.2	(g) 16.9	(h) 13.5	(i) 15.7	(j) 18.5	(k) 14.8	(l) 17.5	(m) 13.9	(n) 16.4	(o) 4.6	(p) 20

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.3	10.2	17.3	13.3	9.6	16.0	12.5	9.1	15.2	44.2	18.5	41.4	17.4	38.9	16.5	19.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 14.2	(b) 12.3	(c) 11.1	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
(a) 14.2	(b) 12.3	(c) 11.1	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-1.1	-13.1	-11.0	-12.2	-4.1	2.5	7.8	12.7	11.7	7.4	0.8	-6.5	-10.2
	DBStd	10.95	9.46	8.18	7.00	5.73	5.25	6.59	2.61	2.73	3.19	4.17	5.42	7.09
	HDD10.0	4261	716	588	688	422	234	98	6	14	87	286	494	627
	HDD18.3	7104	974	821	946	672	491	316	177	205	328	544	744	885
	CDD10.0	198	0	0	0	0	2	32	88	67	9	0	0	0
	CDD18.3	1	0	0	0	0	0	0	1	0	0	0	0	0
	CDH23.3	5	0	0	0	0	0	1	4	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	4.8	5.5	5.8	5.1	4.2	3.5	3.7	4.4	3.8	4.4	5.3	5.8	6.0
Precipitation	PrecAvg	522	34	33	27	22	26	34	63	84	67	53	42	37
	PrecMax	687	76	112	78	51	50	56	122	167	124	136	90	78
	PrecMin	333	8	8	5	7	8	15	10	23	29	16	9	15
	PrecStd	88	17	21	17	12	11	12	27	27	26	28	22	17
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.1	3.4	3.4	9.4	15.1	21.0	23.0	20.6	16.1	11.3	4.8	5.5
		MCWB	1.9	1.5	1.7	4.4	10.6	14.5	17.0	16.0	12.6	7.8	3.1	2.6
	2%	DB	2.7	2.3	1.8	6.4	12.4	18.4	20.2	18.7	13.9	9.3	3.0	2.3
		MCWB	1.3	0.8	0.4	3.6	8.4	13.4	15.4	15.1	11.1	6.9	1.5	0.7
	5%	DB	1.2	1.2	0.1	4.3	10.8	16.5	18.7	17.3	12.8	7.8	1.6	0.9
		MCWB	0.0	0.0	-0.9	2.2	7.4	12.6	14.9	14.0	10.5	6.0	0.7	-0.2
	10%	DB	-1.0	-0.2	-1.4	2.8	9.0	14.8	17.4	16.0	12.1	6.6	0.4	-1.0
		MCWB	-1.9	-1.2	-2.2	1.1	6.4	11.6	13.9	13.4	10.0	5.0	-0.4	-1.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.2	1.8	1.8	5.2	10.7	15.8	18.1	16.8	13.1	8.2	3.7	3.4
		MCDB	3.8	3.1	3.1	8.1	14.4	19.5	22.0	19.5	15.2	10.1	4.6	4.3
	2%	WB	1.3	0.8	0.3	3.7	9.1	14.2	16.1	15.4	11.5	7.3	1.8	0.7
		MCDB	2.7	1.9	1.5	6.5	11.7	17.4	18.9	17.8	13.1	9.3	2.8	2.1
	5%	WB	0.0	0.0	-0.9	2.4	7.6	12.9	15.2	14.5	10.6	6.3	0.8	-0.3
		MCDB	1.2	1.1	0.0	4.4	10.2	15.8	18.0	16.8	12.6	7.9	1.6	0.7
	10%	WB	-1.9	-1.2	-2.3	1.2	6.5	11.8	14.2	13.6	10.0	5.1	-0.4	-1.8
		MCDB	-1.2	-0.2	-1.5	2.7	8.9	14.5	16.9	15.7	11.8	6.6	0.4	-1.0
Mean Daily Temperature Range	5% DB	MDBR	5.3	5.9	6.4	6.6	6.8	7.0	5.7	5.1	4.6	3.8	4.3	5.1
		MCDBR	6.3	5.4	5.6	7.6	8.9	8.7	8.6	7.2	5.7	4.8	4.2	6.3
		MCWBR	5.5	4.6	4.7	5.4	5.5	4.7	4.9	4.3	3.8	3.6	3.8	5.6
	5% WB	MCDBR	6.3	5.9	5.4	7.7	8.0	7.8	7.7	6.5	5.3	4.9	4.1	6.1
		MCWBR	5.5	5.1	4.6	5.7	5.1	4.6	4.7	4.1	3.6	3.8	3.7	5.5
Clear-Sky Solar Irradiance	taub	0.194	0.265	0.268	0.340	0.362	0.365	0.364	0.384	0.318	0.285	0.206	0.184	
	taud	1.970	2.133	2.236	2.081	2.236	2.385	2.444	2.369	2.539	2.389	1.985	1.892	
	Ebn at Noon	426	661	837	827	846	854	845	785	788	653	385	194	
	Edh at Noon	30	63	85	126	119	105	97	96	66	49	28	14	
All-Sky Solar Radiation	RadAvg	0.21	0.86	2.62	4.44	5.22	5.38	4.37	3.40	2.14	0.93	0.26	0.08	
	RadStd	0.03	0.12	0.26	0.30	0.51	0.51	0.28	0.43	0.20	0.11	0.05	0.01	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (13 neighbors)	+0.83	N/A	N/A	N/A	N/A	+0.48	+0.55	N/A	N/A	+35	N/A

Nomenclature: See separate page